
Practical -7

Aim:

To Implement of making a change problem using dynamic programming

Theory:

The Change-Making Problem (often called the Coin Change Problem) aims to find the minimum number of coins needed to make a specific target amount using a given set of coin denominations.

Algorithm:

Step 1: Start.

Step 2: Read the number of coin denominations n .

Step 3: Read the n coin denominations into an array `coins[]`.

Step 4: Read the amount A for which change is required.

Step 5: Create a DP array `dp[0...A]`.

Step 6: Initialize:

because zero coins are required to make amount 0.

Set all other values to infinity.

Step 7: For every amount i from 1 to A :

- Check every available coin.
- If $\text{coin} \leq i$, update:

Step 8: After completing the DP table:

- If `dp[A]` is infinity, the change cannot be made.

- Otherwise, $dp[A]$ gives the minimum number of coins.

Step 9: Display the minimum number of coins.

Step 10: Stop.

Program Code

```
#include <iostream>

#include <vector>

#include <climits>

#include <algorithm>

using namespace std;

int main() {

    int n, amount;

    // Input number of coins

    cout << "Enter number of coin denominations: ";

    cin >> n;

    vector<int> coins(n);

    // Input coin denominations
```

```
cout << "Enter coin denominations: ";

for (int i = 0; i < n; i++) {

    cin >> coins[i];

}

// Input amount

cout << "Enter amount: ";

cin >> amount;

// DP array

vector<int> dp(amount + 1, INT_MAX);

// Base case

dp[0] = 0;

// Calculate minimum coins

for (int i = 1; i <= amount; i++) {

    for (int j = 0; j < n; j++) {

        if (coins[j] <= i && dp[i - coins[j]] != INT_MAX) {
```

```
dp[i] = min(dp[i],  
            dp[i - coins[j]] + 1);  
    }  
}  
}
```

```
// Display result
```

```
if (dp[amount] == INT_MAX) {  
    cout << "Change cannot be made." << endl;  
}  
else {  
    cout << "Minimum number of coins required = "  
        << dp[amount] << endl;  
}  
cout<<"Sangemvijayendrareddy\n"  
; cout<<"92460118132\n";  
cout<<"5-EN18\n";  
return 0;  
}
```

Result :

```
C:\Users\Admin\Downloads\d X + v
Enter number of coin denominations: 5
Enter coin denominations: 4
6
6
7
9
Enter amount: 100
Minimum number of coins required = 12
Sangemvijayendrareddy
92460118132
5-EN18

Process returned 0 (0x0)   execution time : 52.508 s
Press any key to continue.
```

